

Contention-Aware Joint Routing and Spectrum Allocation for Quantum Entanglement Distribution

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Abstract—Quantum entanglement distribution over optical networks requires transmitting both photons of an EPR pair along edge-disjoint paths from a single entanglement source to consumer node pairs. Prior work separates routing from spectrum allocation and assumes exclusive use of spectrum slots by consumer pairs and freely tunable channel widths, which are impractical and resource-inefficient under realistic fixed slot granularity. In this paper, we formulate the contention-aware joint Routing and Spectrum Allocation (RSA) problem as an ILP that maximizes the minimum EPR pair throughput across all consumer pairs. To scale beyond full path enumeration, we propose the Random Clique Heuristic (RCH), which identifies contention-prone demand pairs via a conflict graph and generates detour candidates through stochastic soft weight inflation. The resulting ILP is solved via Lagrangian relaxation with a greedy primal heuristic. Experiments on a realistic topology demonstrate that our approach achieves significantly higher minimum throughput than the separated RSA baseline with practical computational cost.

Index Terms—Quantum networking, Entanglement distribution, Routing and Spectrum Allocation (RSA).

I. INTRODUCTION

Quantum networks are emerging as critical infrastructure for Quantum Key Distribution and distributed quantum computing. The no-cloning theorem, which states that it is impossible to perfectly copy qubits, makes quantum communication theoretically secure, while the quantum computing infrastructure is considered a key next-generation computing paradigm due to the quantum speedup. For any use cases, the transmission of qubits is the critical foundation. In particular, the entanglement-based communication based on entangled qubits (e.g., EPR pairs) distinguishes quantum communication from classical communication.

Hence, the most fundamental building block of quantum networks is the preparation and distribution of entangled qubits. A common and practical method to generate a pair of entangled qubits is Spontaneous Parametric Down-Conversion (SPDC), where a single photon is split into two correlated photons when a high-energy pump beam is directed toward a nonlinear crystal [1]. Since the source of entanglement pairs is a single device with the nonlinear crystal, we need to transmit the two photons to the respective consumer nodes between which entanglement-based communication happens. This phase is called *entanglement distribution*. To further extend the connectivity, entanglement swapping can be applied to two pairs of entangled qubits at a quantum repeater. However,

unfortunately, quantum repeater technologies have not yet met the maturity for actual deployment as a network device. Hence, this paper focuses on the entanglement distribution from an entanglement generation source to consumer node pairs in a *repeaterless* quantum network, which is crucial for concurrent and near-future deployment of intermediate-size quantum networks and will be the foundation for large-scale repeater-based quantum networks.

[2]

Bali et al. [3] established the RSA framework for entanglement distribution over optical networks in their pioneering work on a quantum network resource allocation problem. While their network modeling and analysis lay a solid foundation for all quantum resource allocation problems, they put a strong assumption that each spectrum channel is occupied by only one node pair, regardless of routing. This assumption, combined with a freely tunable channel width, enables them to simplify their solution by separating routing from spectrum allocation. The main contribution of this paper is to relax the exclusive use of spectrum channels by consumer pairs to a more practical and resource-efficient disjointness constraint that allows two distinct consumer pairs to share the same spectrum channel unless their paths for entanglement distribution overlap. Also, we assume a fixed slot granularity, following the flex-grid standard in classical optical networks. This realistic assumption poses a new constraint on the number of distribution paths on each fiber link, while the prior work solved the RSA problem with the guarantee that each fiber always has at least one channel slot for every demand pair because of their tunable channel width. This capacity constraint requires a joint optimization of routing and spectrum allocation, since the separate approach could produce an infeasible solution when a given route does not have sufficient capacity.

The main contributions of this paper are as follows. First, we formulate the contention-aware joint RSA problem as an ILP that maximizes the minimum EPR pair throughput across all consumer pairs. As discussed, our problem allows the repeated use of a spectrum slot among disjoint paths and limits the number of paths hosted on each fiber based on the predefined spectrum slot width. To handle the bilinear coupling between routing and channel assignment variables, we apply the McCormick envelope technique to obtain a tractable ILP formulation. Second, we propose the Random Clique Heuristic

(RCH), a scalable path augmentation algorithm that identifies the most contention-prone demand pairs via a conflict graph and increases their routing candidates through stochastic soft weight inflation. This allows the ILP to consider detour paths precisely where they are most needed, without enumerating all possible path pairs. Third, we solve the resulting ILP efficiently using Lagrangian relaxation, where the disjointness constraint is dualized and minimized by a subgradient algorithm with our greedy primal heuristic to recover feasible lower bounds at each iteration. Experiments on a realistic topology (ILEC) demonstrate that our approach significantly outperforms the separated RSA baseline. We also analyze the impact of a few key hyperparameters of our heuristic approaches to achieve the best balance between solution quality and computational cost.

II. THE JOINT ROUTING AND SPECTRUM ALLOCATION PROBLEM

A. A Motivating Example

To motivate the need for our contention-aware joint RSA problem, let us consider a topology in Fig. 1. In the topology, two fibers, (0, 1) and (1, 2), are remarkably shorter than the others, (0, 2), (0, 3), and (0, 4). Node 0 is the source node, while Nodes 1 to 4 are consumer nodes that demand EPR pairs between every pair: $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$.

When we apply a two-stage solution similar to the work in [3], where the routing is given by the shortest path algorithm separately from the spectrum allocation, all five demand pairs except for (3, 4) will select a path traversing fiber (0, 1) because a path $0 \rightarrow 1 \rightarrow 2$ is the shortest path from the source to Node 2. This routing is feasible only if there is a sufficient number of channels (5 channels in this case) on the busy fiber (0, 1) to provide the spectrum-level disjoint paths, which is not the case in general.

Since the routing and spectrum allocation are solved separately, there is no feasible solution in the spectrum allocation when the number of channels is less than the number of consumer pairs using busy fibers. Thus, we define a joint optimization problem of routing and spectrum allocation, where alternative paths are considered for such contention scenarios.

Table I presents the result of the two-stage optimization as Shortest Pair Baseline. As discussed, the consumer node pairs (2, 3) and (2, 4) insist to use the paths that go through an edge (0, 1). As a result, five consumer node pairs try to use (0, 1), and it cannot provide a feasible solution when the number of channels is 3.

B. The Problem

We consider the standard entanglement distribution setup using an optical network technology. There is a single entanglement source, which generates EPR pairs for consumer nodes. Both photons in an EPR pair need to be transmitted in the same spectrum channel. Thus, a consumer node pair needs to have a pair $R = (p_i, p_j)$ of edge-disjoint paths from the source node to receive the EPR pairs.

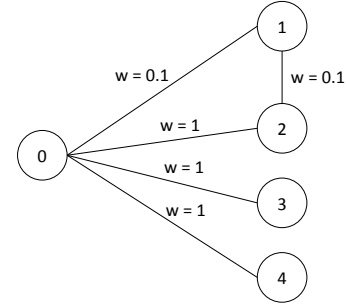


Fig. 1: A Motivating Example. w represents the length of the fiber cable.

We define two variables, $Y_{R,(i,j)}$ and $X_{x,R}$, that indicate the routing decision and spectrum allocation, respectively. These two variables collectively determine which channel on which paths will be used to transmit EPR pairs to a given consumer pair.

- $Y_{R,(i,j)} = 1$ if a consumer node pair (i, j) uses a path pair R from the source s .
- $X_{x,R} = 1$ if a channel x of a path pair R is used.

The transmittance η_R is a function of the total length $l_{(i,j)}$ of the paths in the pair $R = (p_i, p_j)$:

$$\eta_R = 10^{-l_{(i,j)}/10}, \quad (1)$$

where $l_{(i,j)} := \sum_{e \in E(p_i)} w(e) + \sum_{e \in E(p_j)} w(e)$, when $E(p)$ represents a set of edges in a given path p . Then, the total EPR distribution rate from a source s to a consumer node pair (i, j) is

$$\sum_x \sum_R X_{x,R} Y_{R,(i,j)} \eta_R. \quad (2)$$

Note that our formulation assumes the uniform source rate across different frequencies. The different source rates depending on frequency can be easily added to Eq. (2) by weighting the transmittance by a predefined constant C_x that represents the source generation rate per channel. We will hold the uniform generation rate assumption in this paper, since these weight constants do not fundamentally change the problem complexity.

We chose the path-level formulation, where candidate paths are given in the problem input, instead of the edge-level formulation, where a path is constructed in the optimization by boolean variables that indicate the use of edges. This path-level formulation prevents edge-level fragmentations, which makes it difficult to find valid paths that preserve the continuity constraint. Suppose $\mathcal{R} = \{R\}$ represents a set of all pairs of disjoint paths from s to all consumer node pairs. Then, the contention-aware joint RSA problem is defined as a maximization problem of the lowest EPR pair generation rate

assigned to a consumer pair as follows.

$$\text{Maximize } T \quad (3)$$

$$\sum_x X_{x,R} \leq \left\lfloor \frac{W}{W_c} \right\rfloor \quad \forall R \quad (4)$$

$$(X_{x,R} + X_{x,R'}) D_{R,R'} \leq 1 \quad \forall x, \forall R, \forall R' \quad (5)$$

$$\sum_R Y_{R,(i,j)} = 1 \quad \forall (i,j) \quad (6)$$

$$\sum_x \sum_R X_{x,R} Y_{R,(i,j)} \eta_R \geq T \quad \forall (i,j) \quad (7)$$

Eq. (4) is a constraint to provide the contention awareness. While there is no standard that defines the channel width in quantum networks yet, we assume the minimum slot granularity of 6.25 GHz, following the flex-grid standard. This means that the number of available channels is bounded by the total available spectrum band W and the channel size W_c ($= 6.25$ in our experiments, without loss of generality). Eq. (5) ensures that a channel x is used by only one of the overlapping path pairs. Here, the overlapping of two path pairs is defined as the existence of at least one edge that is used by two paths in the distinct path pairs. If such path pairs use the same channel, the transmission experiences a conflict on the shared edge. To define this constraint, we introduce a precomputed variable $D_{R,R'}$ that indicates the overlap. When two path pairs R and R' overlap, $D_{R,R'} = 1$; otherwise, 0. Given the sets of edges in both path pairs, $D_{R,R'}$ can be precomputed by taking the intersection of them, which is not computationally expensive. Since our problem includes the routing decision, we need to make sure that each consumer node pair (i,j) receives a path pair that can serve the demand, as defined in Eq. (6). The constraint in Eq. (7) sets the lower bound T of the EPR rates discounted by the transmittance based on the distance from the source node. By maximizing the lower bound in the objective function, we maximize the lowest rate among all consumer node pairs.

The rate bound constraint in Eq. (7) is bilinear and cannot be included in an ILP formulation. We use the famous McCormick envelope technique that introduces a proxy variable $Z = X_{x,R} Y_{R,(i,j)}$ and converts it to an ILP problem. This conversion adds three additional constraints for each constraint to represent the logical AND operation:

$$Z \leq X_{x,R}; \quad Z \leq Y_{R,(i,j)}; \quad z \geq X_{x,R} + Y_{R,(i,j)} - 1. \quad (8)$$

III. THE CONTENTION-AWARE RANDOM CLIQUE HEURISTIC

The original formulation of the joint RSA problem requires us to decide routing and spectrum allocation from all possible path pairs based on the ILP problem. We call this setup the full enumeration method, where we enumerate all possible simple path pairs and use an ILP solver to obtain a joint solution. This method provides all possible routing options to the ILP and will achieve an optimal solution even when the number of channels is not sufficient. Table I indicates that the full enumeration method can find an optimal solution in

the motivating example in Section II-A. However, this solution obviously does not scale due to the exponential number of path pairs as the size of the topology increases. This motivates us to devise a contention-aware heuristic solution named *Random Clique Heuristic (RCH)*, which augments critical path pairs that experience contentions. Then, we solve the modified ILP problem only with the subset of path pairs constructed by RCH, using the Lagrangian relaxation method.

A key observation we obtained through experiments was that there were a few bottleneck links or paths that experienced contention and reduced the minimum throughput. While the full path enumeration is a sufficient solution, it is not necessary to check all path pairs in practice to improve the throughput. Thus, we focus on the demand pairs that are competing with each other and provide *detour* options for them in the routing, which could be later found by the Lagrangian optimization steps.

A. Seeding via Suurballe's Algorithm

First, RCH finds a minimal set of path pairs that must be included in the routing choices or feasible solution space. We utilize the approach proposed in the prior work [3]. For a demand (i,j) , we seek the minimum-cost pair of edge-disjoint paths (p_i, p_j) from source s to both destinations simultaneously. We reduce this to a single-commodity shortest-path problem by introducing a pseudo-destination t^* with zero-weight edges to i and j , then applying Suurballe's algorithm to find two edge-disjoint shortest paths $s \rightarrow t^*$. Removing t^* yields p_i and p_j , which are edge-disjoint by construction.

B. Conflict Graph and Maximum Clique

Next, we identify the demand pairs that potentially experience resource contention based on an auxiliary graph called the *conflict graph*. Let Δ denote a set of all demands (i,j) . Given the seed pairs, we construct the conflict graph H in which two demands are adjacent if their seed pairs share at least one network edge. For any two demand pairs $d := (i,j)$ and $d' := (i',j')$

$$(d, d') \in E(H) \iff (E(p_i) \cup E(p_j)) \cap (E(p'_i) \cup E(p'_j)) \neq \emptyset, \quad (9)$$

where (p_i, p_j) and (p'_i, p'_j) are the seed path pairs selected for d and d' , respectively.

C. Stochastic Augmentation via Soft Weight Inflation

A maximum clique in H identifies the most mutually constrained demands, which benefit from detour options in the routing. We use the greedy approximation algorithm to obtain the clique [4].

Each demand (i,j) in the maximum clique $\mathcal{K}$ is independently selected for augmentation with probability $p_{\text{clique}} \in (0, 1]$. For each selected demand (i,j) , we generate up to k detour pairs through iterative soft weight inflation:

Let $\bar{w} := \text{median}_{e \in E} w(e)$ be the median edge weight of G , and let $\alpha > 0$ be an inflation factor (a hyperparameter). The perturbed weight is $w_{\text{inf}} = \alpha \bar{w}$.

TABLE I: Shortest-pair baseline vs. full enumeration vs. RCH ($p_{\text{clique}}=0.5$, seed 42) on the tight-cap topology. Shown path pairs are those selected by the ILP solution. Bold entries deviate from the shortest path.

Demand	Shortest Pair Baseline		Enumeration		RCH ($p_{\text{clique}}=0.5$)	
	Path pair ($p_i p_j$)	η	Path pair ($p_i p_j$)	η	Path pair ($p_i p_j$)	η
(1, 2)	(0 $\rightarrow$ 1) (0 $\rightarrow$ 2)	0.7762	(0 $\rightarrow$ 1) (0 $\rightarrow$ 2)	0.7762	(0 $\rightarrow$ 1) (0 $\rightarrow$ 2)	0.7762
(1, 3)	(0 $\rightarrow$ 1) (0 $\rightarrow$ 3)	0.7762	(0 $\rightarrow$ 1) (0 $\rightarrow$ 3)	0.7762	(0 $\rightarrow$ 1) (0 $\rightarrow$ 3)	0.7762
(1, 4)	(0 $\rightarrow$ 1) (0 $\rightarrow$ 4)	0.7762	(0 $\rightarrow$ 1) (0 $\rightarrow$ 4)	0.7762	(0 $\rightarrow$ 2 $\rightarrow$ 1) (0 $\rightarrow$ 4)	0.6166
(2, 3)	(0 $\rightarrow$ 1 $\rightarrow$ 2) (0 $\rightarrow$ 3)	0.7586	(0 $\rightarrow$ 2) (0 $\rightarrow$ 3)	0.6310	(0 $\rightarrow$ 2) (0 $\rightarrow$ 3)	0.6310
(2, 4)	(0 $\rightarrow$ 1 $\rightarrow$ 2) (0 $\rightarrow$ 4)	0.7586	(0 $\rightarrow$ 2) (0 $\rightarrow$ 4)	0.6310	(0 $\rightarrow$ 1 $\rightarrow$ 2) (0 $\rightarrow$ 4)	0.7586
(3, 4)	(0 $\rightarrow$ 3) (0 $\rightarrow$ 4)	0.6310	(0 $\rightarrow$ 3) (0 $\rightarrow$ 4)	0.6310	(0 $\rightarrow$ 3) (0 $\rightarrow$ 4)	0.6310
T	0.0000 (infeasible with 3 channels)		0.6310		0.6166	

At each iteration $t = 1, \dots, k$:

- 1) Collect the edge set $\mathcal{E}_t$ of *all* path pairs found so far for demand (i, j) (seed pair plus any prior detour pairs):

$$\mathcal{E}_t = \bigcup_{r \leq t} (E(p_i^{(r)}) \cup E(p_j^{(r)})).$$

- 2) Temporarily set $w(e) \leftarrow w_{\text{inf}}$ for all $e \in \mathcal{E}_t$.
- 3) Rerun Suurballe's on the inflated graph to obtain (p'_i, p'_j) .
- 4) Restore all original weights. If (p'_i, p'_j) is not already in $\mathcal{R}_d$, add it as a new candidate pair; otherwise stop early.

By inflating all edges of the previously discovered pairs rather than a single edge, each iteration forces a route that is maximally different from all known candidates, yielding diverse detours in few iterations. The inflation factor α controls the trade-off: larger α more strongly discourages the reuse of known edges. Edge-disjointness of each detour pair is guaranteed by Suurballe's. The full procedure is given in Algorithm 1.

IV. LAGRANGIAN RELAXATION

Given a set of feasible path pairs, including the detour options added by RCH, we can solve the joint optimization of routing and spectrum allocation based on the ILP formulation. For further scalability, we use the Lagrangian relaxation technique to efficiently obtain an optimal solution.

A. Lagrangian Relaxation of the Disjointness Constraint

The disjointness constraint (eq. (5)) couples path pairs through shared channels and is the primary source of computational demand. We add it as a soft constraint to the objective with non-negative Lagrangian multipliers $\lambda_{x,R,R'} \geq 0$ for each conflicting pair (R, R') and channel x . Adding the penalty for each violation $X_{x,R} + X_{x,R'} - 1 > 0$ gives the Lagrangian:

$$\begin{aligned}
L(\lambda) &= \max_{\substack{X,Y,Z,T \\ \text{s.t. eqs. (4)(6)(7)(8)}}} T \\
&+ \sum_{\substack{(R,R'): \\ D_{R,R'}=1}} \sum_{x \in \mathcal{X}} \lambda_{x,R,R'} (1 - X_{x,R} - X_{x,R'}) \quad (10) \\
&= \max_{\substack{X,Y,Z,T \\ \text{s.t. eqs. (4)(6)(7)(8)}}} \left[T - \sum_{R \in \mathcal{R}} \sum_{x \in \mathcal{X}} p_{x,R} X_{x,R} \right] \\
&+ \sum_{\substack{(R,R') \\ x}} \lambda_{x,R,R'}, \quad (11)
\end{aligned}$$

Algorithm 1 Random Clique Heuristic (RCH)

Input: Graph G , source s , demand set $\mathcal{D}$, probability p_{clique} , inflation factor $\alpha > 0$, augmentation count $k \geq 1$

Output: Candidate path pair sets $\{\mathcal{R}_d\}_{d \in \mathcal{D}}$

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1: // Stage 1: Seeding via Suurballe's
2: for each demand  $d = (i, j) \in \mathcal{D}$  do
3:    $(p_i, p_j) \leftarrow \text{SUURBALLE}(G, s, i, j)$ 
4:    $\mathcal{R}_d \leftarrow \{(p_i, p_j)\}$ 
5: end for
6: // Stage 2: Conflict graph and approximate max clique
7: Build conflict graph  $H$  over  $\{\mathcal{R}_d\}_{d \in \mathcal{D}}$ 
8:  $\mathcal{K} \leftarrow \text{APPROXMAXCLIQUE}(H)$ 
9: // Stage 3: Stochastic soft weight inflation augmentation
10:  $\bar{w} \leftarrow \text{median}_{e \in E} w(e)$ ;  $w_{\text{inf}} \leftarrow \alpha \cdot \bar{w}$ 
11: for each demand  $d = (i, j) \in \mathcal{K}$  do
12:   Draw  $u \sim \text{Uniform}(0, 1)$ 
13:   if  $u < p_{\text{clique}}$  then
14:     for  $t = 1, \dots, k$  do
15:        $\mathcal{E}_t \leftarrow \bigcup_{r \in \mathcal{R}_d} (E(p_i^{(r)}) \cup E(p_j^{(r)}))$ 
16:        $w(e) \leftarrow w_{\text{inf}}$  for all  $e \in \mathcal{E}_t$ 
17:        $(p'_i, p'_j) \leftarrow \text{SUURBALLE}(G, s, i, j)$ 
18:       Restore  $w(e)$  for all  $e \in \mathcal{E}_t$ 
19:       if  $(p'_i, p'_j) \notin \mathcal{R}_d$  then
20:          $\mathcal{R}_d \leftarrow \mathcal{R}_d \cup \{(p'_i, p'_j)\}$ 
21:       else
22:         break
23:       end if
24:     end for
25:   end if
26: end for
27: return  $\{\mathcal{R}_d\}_{d \in \mathcal{D}}$ 

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where the aggregated penalty coefficient for (x, R) is

$$p_{x,R} = \sum_{R': D_{R,R'}=1} (\lambda_{x,R,R'} + \lambda_{x,R',R}). \quad (12)$$

For any $\lambda \geq 0$, $L(\lambda)$ is a valid upper bound on the true optimal T^* . The Lagrangian dual problem minimizes this bound:

$$L^* = \min_{\lambda \geq 0} L(\lambda) \geq T^*. \quad (13)$$

For a fixed λ , the inner maximization in eq. (11) is an ILP with constraint eqs. (4), (6), (7), and (8), which is drastically smaller than the original ILP problem.

B. Subgradient Algorithm

We minimize $L(\lambda)$ via the subgradient method. At iteration k with multipliers $\lambda^{(k)}$:

- 1) Compute penalties $p^{(k)}$ based on eq. (12) and solve the relaxed LP problem to obtain $X^{(k)}$, $Y^{(k)}$, $T^{(k)}$, and the Lagrangian bound $UB^{(k)}$.
- 2) Apply a greedy primal heuristic (See Section IV-C) to $Y^{(k)}$ to obtain a feasible integer solution with throughput $LB^{(k)}$. Then, update LB with $LB = \max(LB, LB^{(k)})$.
- 3) Compute the subgradient of $L(\lambda)$ with respect to $\lambda_{x,R,R'}$:

$$g_{x,R,R'}^{(k)} = X_{x,R}^{(k)} + X_{x,R'}^{(k)} - 1. \quad (14)$$

Here, a positive value indicates a violated disjointness constraint.

- 4) Update multipliers with the Polyak step:

$$\lambda_{x,R,R'}^{(k+1)} = \max\left(0, \lambda_{x,R,R'}^{(k)} + \alpha^{(k)} g_{x,R,R'}^{(k)}\right), \quad (15)$$

where the step size is

$$\alpha^{(k)} = \frac{\theta^{(k)} (UB^{(k)} - LB)}{\|g^{(k)}\|^2}, \quad (16)$$

and $\theta^{(k)} \in (0, 2]$ is halved every 20 iterations without improvement in LB (Polyak schedule).

The algorithm terminates when the optimality gap $(UB - LB)/(1 + |LB|) < \epsilon$, or when iteration budgets are exhausted.

C. Greedy Primal Heuristic

To obtain a lower bound at each iteration, we round the LP solution to a feasible integer solution. First, our greedy primal heuristic assigns exactly one channel to each chosen path pair. When the chromatic number of the conflict graph is less than or equal to the number of the channels ($|\mathcal{X}|$ -colorable), this encourages producing a $T > 0$ solution. Note that because the transmittance is a function of distance, T may become negligibly small depending on the routing. Then, the heuristic adds channels in round-robin order iteratively until no further conflict-free assignment is possible. We sorted the path pairs based on η so that weaker pairs will receive the additional path pairs first in the iterations.

V. EVALUATION EXPERIMENTS

We conducted our experiments in the small motivating example and the ILEC topology with 17 nodes in the prior work [3]. Without loss of generality, we designated node A to be the source node in the following experiments. While a different source node will change the max-min throughput in the result, the fundamental problem complexity does not change. Also, in the ILEC graph, 15 nodes, including the source node A, among the 17 nodes are fully connected. Therefore, it will provide a fairly reasonable number of disjoint path pairs. Note that this high-density network topology is *not* a special assumption. In many practical, intermediate-size quantum networks, we assume a full-mesh or dense network topology. This is mostly because the transmission of photos

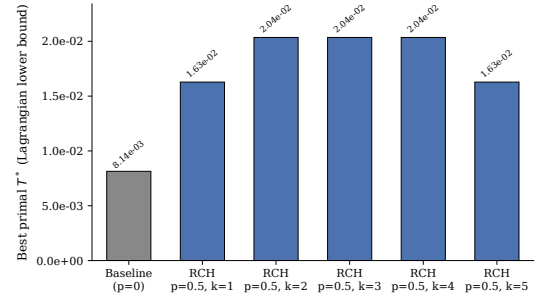


Fig. 2: The resulting minimum throughput T (normalized) under varying RCH parameters.

over longer distances and multiple hops is not yet an easy task in the practical quantum networks. While the prior work assumes different EPR pair generation rates at a source, we assume that all channels generate EPR pairs at the same rate. However, the extension is straightforward by adding a constant weight vector in the throughput calculation in eq. (7), similar to the prior work. We did not include the channel rates because they do not change the complexity of the ILP problem.

As a baseline for performance comparison, we will use the prior work's RSA solution that separately solves routing and spectrum allocation. When we set our hyperparameter $p_{\text{clique}} = 0$, there will be no detour options added for the subsequent ILP (See Section III-C). Therefore, it is equivalent to the separate routing process by Suurballe's algorithm.

Experiments were conducted on a desktop workstation equipped with a 13th Gen Intel Core i9-13900KF processor (24 cores, up to 5.8 GHz), an NVIDIA GeForce RTX 4080 GPU (16 GB GDDR6X), 64 GB DDR5 RAM (5200 MHz), and a 1 TB NVMe M.2 PCIe SSD. The ILP and LP were solved using Gurobi 13.0.1 via the gurobipy Python interface.

Figure 2 illustrates the obtained objective function, which is the minimum throughput among all demand pairs. p represents the percentage of demand pairs selected in the augmentation (p_{clique} in Algorithm 1). k is the maximum number of detour pairs that can be added. Our solutions (in blue) clearly outperform the separated RSA solution (in grey) because our solutions can utilize detour path pairs to avoid excessive contention, especially when the spectrum allocation is tight due to the resource availability. When comparing different hyperparameters, $k = 2$ provides a better lower bound than $k = 1$. This is because more detour options are available to the ILP when k is larger. However, the impact of detour options seems to fade after $k = 2$, probably because other detours are too long, which negatively impact the transmittance, or simply do not exist. The performance deterioration in $k = 5$ could be due to the computation budget constraint. As will be discussed, the ILP solver process was terminated based on the iteration budget (100 iterations) in the $k = 5$ case.

Figure 3 shows the link utilization (PDFs) under varying RCH hyperparameter k . Compared to the baseline approach ($p_{\text{clique}} = 0$), our approaches remarkably increased the utilization of the spectrum channels of many links. Also, it is

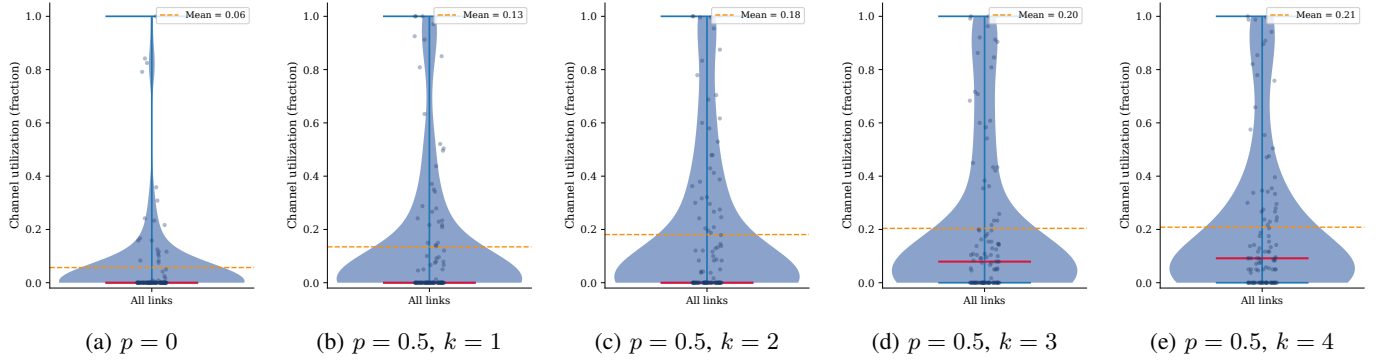


Fig. 3: Utilization PDFs under varying RCH parameters.

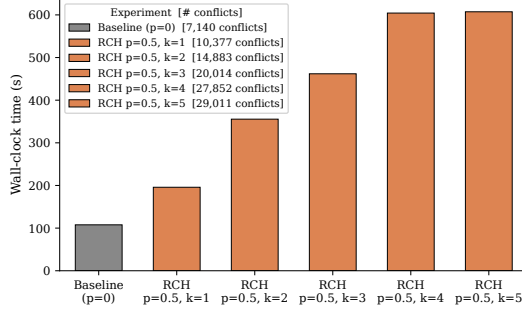


Fig. 4: Wall-clock time in seconds under varying RCH parameters.

observed that the number of highly utilized links (blue dots near utilization 1.0) increases in our approach, while there are very few highly utilized links with the baseline approach. This means that detour path pairs enabled offloading some demands to secondary links until their saturation to improve the minimum throughput.

Figure 4 summarizes the wall-clock time of the ILP solver while solving the Lagrangian problem. We set the iteration budget limit to 100 iterations, which was reached by the $k = 4$ and 5 cases. The number of conflicts in the legend represents the number of demand pairs that share at least one edge ($D_{R,R'} = 1$), which each add a constraint eq. (5).

VI. CONCLUSION

In this paper, we addressed the contention-aware joint Routing and Spectrum Allocation (RSA) problem for quantum entanglement distribution over an optical network. We demonstrated that separating routing and spectrum allocation can result in infeasible or suboptimal solutions when spectrum resources are limited and multiple demand pairs compete over shared fibers. To address this, we formulated the joint RSA problem as an ILP with a max-min throughput objective and proposed the Random Clique Heuristic (RCH) to scalably augment routing candidates for the most contention-prone demand pairs. By combining RCH with Lagrangian relaxation and a greedy primal heuristic, our approach achieves significantly higher minimum throughput than the separated baseline

while maintaining practical computational time. Experimental results on both the motivating example and a realistic topology confirm that RCH with moderate augmentation depth strikes the best balance between solution quality and computational overhead, with diminishing returns.

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